RESEARCH PROJECT HANDOVER

Beyond Individual Risk Factors: Using Explainable Machine Learning to Identify Child Profiles Associated with Foundational Learning in Jamaica

Project status: Research design and preliminary data exploration
Data source: Jamaica Multiple Indicator Cluster Survey (MICS), 2022
Primary dataset: `fs.sav`
Programming language: Python (preferred), with R as an option for complex survey analysis
Reference methodology: Alvarez-Garcia et al. (2024), *Uncovering Student Profiles: An Explainable Cluster Analysis Approach to PISA 2022*

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1. Research Overview

1.1 Research Abstract

This study investigates how children's individual and household circumstances intersect with foundational reading and numeracy outcomes among Grade 2 and 3 children in Jamaica.

Specifically, it asks:

Can explainable machine learning identify distinct profiles of children based on their lived circumstances, and how are these profiles associated with foundational learning proficiency?

Drawing upon established explainable cluster analysis methods, this study uses data from the 2022 Jamaica Multiple Indicator Cluster Survey to identify child profiles using sparse principal component analysis, multiple correspondence analysis, and k-means clustering. Cluster membership is then compared with foundational learning outcomes to determine which combinations of circumstances are associated with learning vulnerability.

The findings support targeted, cross-sector policies by identifying groups of children experiencing intersecting disadvantages rather than isolated risk factors. The study contributes evidence toward Sustainable Development Goal indicator 4.1.1 and advances a child-rights perspective by linking foundational learning with rights to education, health, and protection.

1.2 Central Research Question

Can explainable machine learning identify distinct profiles of Grade 2 and 3 children in Jamaica based on their individual and household circumstances, and how are these profiles associated with foundational reading and numeracy proficiency?

1.3 Research Objectives

The study has four main objectives:

1. Identify distinct profiles of Grade 2 and 3 children based on combinations of individual, household, educational, and environmental circumstances.
2. Characterize and explain these profiles using interpretable machine-learning methods.
3. Examine differences in foundational reading and numeracy proficiency across the identified profiles.
4. Interpret the findings in relation to intersecting disadvantages, children's rights, and policies aimed at improving foundational learning in Jamaica.

1.4 Conceptual Motivation

Most studies of foundational learning inequalities examine individual predictors, such as household wealth, sex, disability, or parental education.

This study takes a different approach.

Rather than primarily estimating the independent association between each characteristic and learning outcomes, it seeks to identify groups of children who share particular combinations of circumstances.

For example, children from households with similar socioeconomic resources may differ in parental educational involvement, functional difficulties, or exposure to disciplinary practices.

These combinations may be associated with different patterns of foundational learning proficiency.

The objective is to identify meaningful profiles of children's lived circumstances that could inform more targeted and coordinated policy responses.

Important methodological principle: Reading and numeracy proficiency should not be included among the variables used to construct the clusters.

Instead, profiles should be identified independently using contextual characteristics and subsequently compared on foundational learning outcomes.

This ensures that observed differences in proficiency are not built directly into the clustering procedure.

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2. Data Source and Available Files

The study uses the **2022 Jamaica Multiple Indicator Cluster Survey (MICS)**.

The following datasets are available:

File	Description	Proposed role
`fs.sav`	Children aged 5–17 years	Primary analytical dataset
`hl.sav`	Household members	Optional supplementary dataset
`hh.sav`	Households	Optional supplementary dataset
`Readme_Jamaica_MICS6.rtf`	Survey data documentation	Dataset reference

The researcher also has access to a PDF of the reference paper:

Uncovering Student Profiles: An Explainable Cluster Analysis Approach to PISA 2022.

2.1 Decision Regarding Dataset Selection

The initial methodological discussion considered merging `fs.sav`, `hl.sav`, and `hh.sav`.

However, subsequent examination of the variable labels in `fs.sav` showed that this file already contains many relevant individual, family, educational, and socioeconomic indicators.

Therefore, the current recommendation is:

Begin with `fs.sav` as the sole analytical dataset.

Only consider merging additional files if important theoretically relevant variables are unavailable in `fs.sav`.

This approach should simplify data preparation and reduce the risk of unnecessary merging errors.

The decision is provisional and should be reassessed after a complete variable audit.

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3. Previous Data Exploration

Some preliminary Python code was developed to explore the `fs.sav` dataset.

The dataset was loaded using:

```
import pyreadstat

df, meta = pyreadstat.read_sav("fs.sav")

meta.column_names_to_labels
```

The researcher also identified the following major groups of variables:

- Child demographic characteristics.
- Educational participation and schooling history.
- Health and health insurance.
- Child labour and household responsibilities.
- Child discipline.
- Child functioning and disability.
- Parental involvement in education.
- Foundational reading assessment.
- Foundational numeracy assessment.
- Household socioeconomic characteristics.
- Survey design variables.

The researcher has already produced a variable-label dictionary for `fs.sav`.

However, a comprehensive analysis of missingness, category frequencies, and eligibility has not yet been completed.

No final clustering dataset has been established.

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4. Target Population and Sample Selection

4.1 Intended Study Population

The intended analytical population is children attending **Grade 2 or Grade 3** during the 2022 Jamaica MICS survey.

The original code used the following variables:

Variable	Description
`CB7`	Attended school or early childhood programme during the current school year
`CB8A`	Level of education attended during the current school year
`CB8B`	Grade attended during the current school year
`FL28`	Result of interview with child

The previous filtering approach was:

```
PRIMARY_LEVEL_CODE = 1
ATTENDING_CODE = 1
COMPLETED_FL_CODE = 1

grade23_df = analysis_df.loc[
    (analysis_df["CB7"] == ATTENDING_CODE)
    & (analysis_df["CB8A"] == PRIMARY_LEVEL_CODE)
    & (analysis_df["CB8B"].isin([2, 3]))
    & (analysis_df["FL28"] == COMPLETED_FL_CODE)
].copy()
```

Important: Codes require verification

The previous code assumed that:

- `CB7 == 1` means currently attending school.
- `CB8A == 1` means primary education.
- `CB8B` values 2 and 3 correspond to Grades 2 and 3.
- `FL28 == 1` means a completed foundational learning assessment.

These assumptions must be verified using the SPSS value labels and survey documentation.

The final sample should not be constructed using unverified coding assumptions.

4.2 Recommended Improvement

Separate the selection process into two stages.

Stage A: Identify all eligible Grade 2 and 3 children.

Stage B: Identify which of these children completed the foundational learning assessment.

This will allow the researcher to report:

- Total number of Grade 2–3 children.
- Number and percentage completing the assessment.
- Number with valid reading outcomes.
- Number with valid numeracy outcomes.
- Differences between assessed and non-assessed children.

The sample size has not yet been confirmed.

This is one of the most important outstanding tasks.

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5. Foundational Learning Outcomes

The study will examine two separate primary outcomes:

1. Foundational reading proficiency.
2. Foundational numeracy proficiency.

These outcomes will be constructed using the foundational learning assessment items in `fs.sav`.

5.1 Reading Assessment Variables

Relevant variables include:

Variable	Description
`FL14`	Practice reading task

FL20A	Total number of words attempted
FL20B	Total number of words incorrect or missed
FL21BA	Reading comprehension question 1
FL21BB	Reading comprehension question 2
FL21BC	Reading comprehension question 3
FL21BD	Reading comprehension question 4
FL21BE	Reading comprehension question 5

The researcher previously attempted to construct a binary reading proficiency variable.

The initial approach classified children as proficient if they passed the decoding threshold and answered all five comprehension questions correctly.

However, **this coding requires revision**.

The standard MICS foundational reading criterion generally involves correctly reading at least 90% of words in a story and correctly answering at least three of five comprehension questions.

The exact Jamaica MICS scoring rules, word count, response codes, and routing instructions must be verified before constructing the final outcome.

In particular, the previous code used:

```
failed_decoding = grade23_df["FL20B"].ge(8)
passed_decoding = grade23_df["FL20B"].lt(8)
```

The cutoff of eight incorrect or missed words should not be accepted without verification.

The code also required all five comprehension responses to equal 1, which may incorrectly classify children who meet the official comprehension threshold.

Recommended action: Reconstruct the reading proficiency indicator from scratch using the official MICS assessment instructions.

5.2 Numeracy Assessment Variables

The researcher previously identified the following numeracy assessment groups:

Variables	Skill assessed

FL23A–FL23F	Number recognition
FL24A–FL24E	Number discrimination
FL25A–FL25E	Addition
FL27A–FL27E	Number patterns

The previous exploratory code checked whether a child answered every numeracy item correctly.

However, this should not automatically be used as the proficiency definition.

The official MICS foundational numeracy indicator uses specific criteria for the assessment tasks.

Recommended action: Verify the official scoring thresholds and construct a binary foundational numeracy proficiency variable accordingly.

5.3 Outcome Coding

The intended final variables are:

foundational_reading
foundational_numeracy

Both should be coded as:

- 1 = Meets the foundational proficiency criterion.
- 0 = Does not meet the criterion.
- Missing = Cannot be classified reliably.

The researcher should document all coding decisions, including how incomplete assessments, refusals, and routing-related missing values are handled.

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6. Potential Variables for Child Profile Construction

The variable-label dictionary indicates that `fs.sav` contains several promising groups of contextual indicators.

These should be treated as candidate variables rather than a finalized feature set.

6.1 Socioeconomic and Demographic Circumstances

Variable	Description
`windex5`	Household wealth quintile
`wscore`	Combined household wealth score
`melevel`	Mother's education
`HH6`	Urban/rural area
`HH7`	Parish
`HL4`	Child's sex
`FSAGE`	Child's age

Household wealth, maternal education, and urban/rural residence are particularly relevant.

However, avoid including both wealth quintile and wealth score in the same clustering specification without justification, because they measure closely related constructs.

Age, sex, and grade may be better treated as supplementary descriptive or adjustment variables rather than primary clustering indicators.

Parish may also be more useful for describing the geographic distribution of profiles than for defining the profiles themselves.

6.2 Home Learning Environment and Parental Involvement

Variable	Description
PR3	Number of children's books or picture books
PR5	Whether the child has ever had homework
PR6	Whether anyone helps the child with homework
PR8	Parent/caregiver attendance at a PTA meeting
PR10	Whether a school report was received
PR11B	Whether parent/caregiver visited school to discuss the child's progress

This domain is potentially one of the most valuable aspects of the project.

It could help identify differences in educational support among children with similar household wealth.

However, some indicators may reflect school practices, child learning needs, or questionnaire routing rather than parental involvement alone.

These distinctions should be considered when interpreting the resulting profiles.

6.3 Child Functioning and Well-Being

Variable	Description
fsdisability	Derived child functional difficulties indicator
FCF19	Difficulty learning things
FCF20	Difficulty remembering things
FCF21	Difficulty concentrating
FCF24	Difficulty making friends
FCF25	Frequency of anxiety or worry
FCF26	Frequency of sadness or depression

These variables could provide a child-rights dimension beyond household socioeconomic conditions.

However, there are important methodological considerations.

First, the derived `fsdisability` variable may overlap with individual functioning questions.

Second, difficulties learning and concentrating may be closely related to educational outcomes.

Third, functional difficulties should not automatically be interpreted as individual deficits or causes of poor performance.

The analysis should emphasize children's circumstances and potential support needs.

A reasonable initial approach is to use the derived functional difficulties indicator, with individual items examined in sensitivity analyses.

6.4 Child Discipline and Protection

Relevant variables include:

Variable	Description
`FCD2C`	Child was shaken
`FCD2D`	Child was shouted or screamed at
`FCD2F`	Child was spanked or slapped
`FCD2G`	Child was hit with an object
`FCD2H`	Child was insulted
`FCD2I`	Child was hit on the face, head, or ears
`FCD2J`	Child was hit on the hand, arm, or leg

These variables may support the study's emphasis on children's rights and protection.

Rather than including every disciplinary item separately, the researcher should consider constructing established composite indicators, such as psychological aggression or physical punishment, where supported by the survey documentation.

The exact coding and eligibility conditions must be verified.

6.5 Digital Access and Educational Circumstances

Variable	Description
CB10D	Participation in online learning during the current school year
CB10EA	Broadband internet access at home
CB10EB	Mobile internet access at home
CB10EC	Internet access through neighbours
CB10ED	Internet access through community hotspots
CB10EE	Other internet access
CB10F	Exclusive access to an available device

These indicators may be especially relevant because the survey was conducted in 2022, following substantial disruption to education during the COVID-19 pandemic.

However, the researcher should examine whether these questions were asked of all relevant children or only those following particular questionnaire routes.

Digital access variables may require careful recoding before inclusion in clustering.

6.6 Additional Variables

Other potentially relevant variables include:

- Child labour and household responsibilities (CL module).
- Health insurance (CB11, fsinsurance).
- School closures and teacher absence (PR12, PR13).
- Schooling history (CB4–CB10).

These should be evaluated based on theoretical relevance and data availability.

The final model should not include every available variable simply because it exists.

The goal is to identify a manageable collection of indicators representing distinct aspects of children's lived circumstances.

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7. Proposed Analytical Methodology

The proposed approach is adapted from Alvarez-Garcia et al. (2024).

The reference study used an explainable cluster analysis framework combining:

1. Data preparation and preprocessing.
2. Dimensionality reduction.
3. Weighted k-means clustering.
4. Supervised classification.
5. SHAP-based cluster explanations.

The proposed Jamaica study adds an explicit analysis of foundational learning outcomes across the resulting profiles.

Stage 1: Data Preparation

Tasks include:

- Load the original SPSS dataset and metadata.
- Verify eligibility and assessment-completion codes.
- Select the Grade 2–3 population.
- Construct reading and numeracy outcomes.
- Audit candidate contextual variables.
- Identify special missing-value codes.
- Examine questionnaire routing.
- Recode variables where appropriate.
- Select a theoretically defensible set of clustering indicators.

Missing data should be handled according to the type and extent of missingness.

Structural missingness caused by questionnaire routing should not automatically be imputed.

Stage 2: Dimensionality Reduction

The proposed methods are:

Multiple Correspondence Analysis (MCA) for categorical variables.

Sparse Principal Component Analysis (SPCA) for suitable continuous variables.

MCA is likely to be particularly important because many MICS indicators are categorical.

SPCA should only be included if the selected variables contain enough suitable continuous indicators to justify its use.

If the data are predominantly categorical, MCA alone may be sufficient.

Factor Analysis of Mixed Data (FAMD) could also be considered as an alternative for mixed numerical and categorical data.

The researcher should avoid forcing both MCA and SPCA into the analysis solely to replicate the reference paper.

Stage 3: Child Profile Identification

Apply k-means clustering to the retained component scores.

Survey-weighted k-means is preferred where feasible.

Initially examine candidate solutions with approximately two to six clusters, adjusting the range according to sample size and observed structure.

Cluster selection should consider:

- Within-cluster sum of squares.
- Elbow plots.
- Silhouette scores.
- Cluster sizes.
- Cluster stability.
- Substantive interpretability.

The number of clusters should not be selected using the elbow method alone.

The final solution should produce meaningful and sufficiently populated child profiles.

Stage 4: Cluster Interpretation

First, describe each cluster using the original contextual variables.

Compare:

- Household wealth.
- Maternal education.
- Home learning resources.
- Parental involvement.
- Child functioning.
- Discipline-related indicators.

- Digital learning circumstances.

Use tables, plots, and standardized comparisons to identify the defining characteristics of each profile.

Assign descriptive profile names based on the observed patterns.

Avoid labeling children as inherently disadvantaged, vulnerable, or low-performing without appropriate evidence.

Stage 5: Explainable Machine Learning

Train a supervised classification model to predict cluster membership from the original contextual indicators.

Potential models include:

- Decision trees.
- Random forests.
- XGBoost.

The reference paper uses XGBoost and SHAP.

SHAP values can be used to identify which variables are most influential in distinguishing each cluster.

The purpose of this model is to explain the clustering solution.

It is not intended to predict reading or numeracy proficiency.

Importantly, SHAP explanations should not be interpreted as causal effects.

Stage 6: Foundational Learning Analysis

After identifying the profiles, compare foundational reading and numeracy proficiency across clusters.

The primary analysis should estimate survey-weighted proficiency rates for each profile.

For each cluster, report:

- Number of sampled children.
- Weighted population share.
- Reading proficiency percentage.
- Numeracy proficiency percentage.

- Confidence intervals.

Additional survey-adjusted logistic regression models may examine differences between profiles after accounting for age and grade.

These analyses should account for the MICS complex survey design.

Stage 7: Robustness and Sensitivity Analyses

Recommended checks include:

- Alternative numbers of clusters.
- Repeated clustering initializations.
- Cluster stability under resampling.
- Alternative contextual variable selections.
- Alternative dimensionality-reduction specifications.
- Sensitivity to missing-data handling.
- Comparison with latent class analysis, if feasible.

Latent class analysis may be especially relevant if the final clustering variables are predominantly categorical.

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8. Survey Design Considerations

The `fs.sav` variable dictionary includes:

Variable	Description
`fsweight`	Children aged 5–17 sample weight
`fshweight`	Children aged 5–17 household sample weight
`PSU`	Primary sampling unit
`stratum`	Survey stratum

The correct weight for the foundational learning analysis must be verified against the MICS documentation.

Survey weights should be considered during clustering, and the appropriate weights, strata, and primary sampling units should be used for population-level estimates and statistical inference.

Ordinary machine-learning methods do not automatically account for complex survey design.

R's `survey` package may be useful for the inferential analysis, even if most preprocessing and clustering are performed in Python.

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9. Issues Identified in the Previous Code

Several issues were identified during review.

Issue 1: Survey design variables were unintentionally excluded

The researcher created a list called `design_columns`, containing variables such as `windex5`, `fsweight`, `PSU`, and `stratum`.

However, this list was not included when constructing `requested_columns`.

As a result, several important contextual and survey design variables were omitted from the earlier `analysis_df`.

Recommendation: Begin with the full `fs.sav` dataset and select variables only after completing the data audit.

Issue 2: Eligibility codes were assumed rather than verified

The previous Grade 2–3 filter assumed particular numerical codes for attendance, education level, and assessment completion.

Recommendation: Verify all codes using SPSS value labels before applying filters.

Issue 3: Reading proficiency was defined too strictly

The previous reading code required all five comprehension questions to be correct.

Recommendation: Reconstruct reading proficiency according to the official MICS criteria.

Issue 4: Numeracy proficiency had not been finalized

The previous code examined whether all numeracy assessment items were correct.

Recommendation: Construct numeracy proficiency using the official task-specific thresholds.

Issue 5: Contextual variable selection was preliminary

Variables were grouped into domains, but their distributions and missingness had not been comprehensively examined.

Recommendation: Complete a variable audit before finalizing the clustering dataset.

Issue 6: Sample size remains unknown

The actual number of eligible Grade 2–3 children has not been established in the handover materials.

Recommendation: Confirm sample size before selecting clustering complexity.

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10. Current Project Status

The project is currently at the **data preparation and methodological planning stage**.

Work completed

- Finalized the research abstract and central research question.
- Identified the 2022 Jamaica MICS as the data source.
- Identified `fs.sav` as the preferred primary dataset.
- Reviewed the methodology of the Alvarez-Garcia et al. reference paper.
- Identified relevant MICS questionnaire modules.
- Developed preliminary Python code for loading and filtering the dataset.
- Explored reading and numeracy assessment items.
- Identified potential contextual variables for clustering.
- Proposed an explainable machine-learning framework.
- Identified methodological issues requiring correction.

Work not yet completed

- Verification of the final Grade 2–3 sample.
- Comprehensive variable audit.
- Official reading proficiency construction.
- Official numeracy proficiency construction.
- Missing-data analysis.
- Final contextual variable selection.

- Dimensionality reduction.
- Clustering.
- Cluster stability assessment.
- XGBoost classification.
- SHAP analysis.
- Survey-weighted learning comparisons.
- Regression analysis.
- Results interpretation.

No empirical child profiles or substantive learning findings have yet been established.

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11. Recommended Next Steps

The incoming researcher should begin from the original `fs.sav` file rather than treating the previous analytical dataset as final.

Priority 1: Verify the Analytical Sample

Load `fs.sav` and inspect the metadata.

Confirm the coding of:

- `CB7`
- `CB8A`
- `CB8B`
- `FL28`

Calculate the total number of eligible Grade 2–3 children and the number completing the foundational learning assessment.

Priority 2: Reconstruct the Learning Outcomes

Use the official Jamaica MICS questionnaire and indicator definitions to construct foundational reading and numeracy proficiency.

Verify all scoring thresholds, missing-value codes, and questionnaire routing conditions.

Priority 3: Conduct a Contextual Variable Audit

For each candidate variable, document:

- Variable name.
- Variable label.
- Questionnaire domain.
- Response categories.
- Number of valid observations.
- Missing percentage.
- Category frequencies.
- Proposed role in the analysis.

Determine which indicators are suitable for clustering.

Priority 4: Finalize the Clustering Methodology

Based on the sample size and variable distributions, determine whether the most appropriate approach is:

- MCA followed by weighted k-means.
- MCA and SPCA followed by weighted k-means.
- FAMD followed by weighted k-means.
- An alternative categorical profiling method.

The chosen approach should be justified by the data rather than by a desire to replicate the reference paper exactly.

Priority 5: Implement Explainable Clustering

Develop the clustering pipeline, evaluate alternative solutions, assess stability, and interpret the resulting profiles.

Use SHAP if a supervised classifier adds meaningful interpretability.

Priority 6: Analyze Foundational Learning

Compare reading and numeracy proficiency across profiles using appropriate survey-adjusted procedures.

Interpret findings as associations rather than causal relationships.

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12. Suggested Project Deliverables

The project could be organized into four reproducible Python notebooks.

Notebook 1: `01_data_audit.ipynb`

Data loading, metadata inspection, sample selection, and contextual variable audit.

Notebook 2: `02_learning_outcomes.ipynb`

Construction and validation of foundational reading and numeracy proficiency.

Notebook 3: `03_child_profiles.ipynb`

Variable preprocessing, dimensionality reduction, clustering, cluster validation, and descriptive profiling.

Notebook 4: `04_explainability_learning.ipynb`

Explainable classification, SHAP analysis, survey-weighted proficiency comparisons, and regression analysis.

A separate R script may be used for complex survey inference if necessary.

Expected outputs include a variable dictionary, documented analytical dataset, cluster diagnostics, profile summaries, SHAP visualizations, proficiency comparison tables, and reproducible analysis code.

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13. Methodological Priorities for the Incoming Researcher

The incoming researcher is encouraged to critically evaluate and improve the proposed methodology.

The following principles should guide the work.

1. Preserve the research question.

The objective is to identify multidimensional child profiles and examine their association with foundational learning.

It is not primarily a prediction study.

2. Avoid outcome leakage.

Reading and numeracy proficiency should not be used to construct the profiles.

3. Prioritize interpretability.

The identified profiles should describe meaningful combinations of children's circumstances.

4. Respect the survey design.

Sampling weights and the complex survey structure must be considered, particularly for population-level inference.

5. Avoid unnecessary model complexity.

The number of variables, dimensions, clusters, and classification models should be appropriate for the available sample.

6. Validate the outcomes carefully.

Incorrect construction of reading or numeracy proficiency would undermine the study regardless of clustering performance.

7. Assess robustness.

The profiles should not depend excessively on arbitrary preprocessing or clustering decisions.

8. Maintain a child-rights perspective.

The analysis should emphasize intersecting circumstances and opportunities for policy support rather than stigmatizing children or families.

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14. Reference Study

Alvarez-Garcia, M., Arenas-Parra, M., & Ibar-Alonso, R. (2024).

Uncovering student profiles. An explainable cluster analysis approach to PISA 2022.

Computers & Education, 223, 105166.

The study provides the primary methodological inspiration.

Its framework combines:

- Data preprocessing and imputation.
- Sparse principal component analysis.
- Multiple correspondence analysis.
- Weighted k-means clustering.
- XGBoost classification.
- SHAP-based explanations.

The Jamaica study should adapt this framework to the structure of MICS and its own research objectives.

The reference paper should be provided to the incoming researcher.

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15. Final Handover Note

The central aim of this project is to move beyond examining isolated correlates of foundational learning and instead identify distinct groups of Jamaican children whose circumstances reflect different combinations of socioeconomic, educational, health-related, family, and protection-related experiences.

The proposed analytical strategy uses explainable clustering to identify these groups and subsequently examines whether foundational reading and numeracy proficiency differ across them.

The available `fs.sav` dataset appears to contain many of the necessary indicators, making a single-dataset approach a reasonable starting point.

However, the analytical sample, outcome construction, contextual variable selection, and final clustering methodology remain to be verified.

The incoming researcher should begin with a systematic audit of the original data, validate the foundational learning indicators, and then determine the most appropriate clustering specification.

The immediate priority is not to run machine-learning models, but to establish a valid, well-documented analytical dataset that can support meaningful and reproducible child-profile analysis.